PyActiveStorage
Preliminary Testing Results

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Version 2: Distributed via Slack

Revision History

Revision	Date	Author(s)	Description
1	25.03.2026	V.S	Initial document created with draft of an overview in Section 1. Tests completed so far filled into section 2. An attempt to interpt the results is undergone in Section 3, currently on hold until the last tests are completed. Distributed via Slack
2	April 17, 2026	V.S	Pyactivestorage vs local, https s3 and some cache testing results, values and info added.

1 Overview

Testing undertaken to demonstrate performance of `PyActiveStorage`, and to quantify factors which impact its ability to perform remote reductions. Different settings are compared and tested, listed below:

- Chunk size
- Network: UoR Eduroam vs UoR Ethernet
- Methods (min max count sum mean) [Not completed, Note: weights for sum and mean]
- Data retrieval: Object `s3` store vs `https` protocol
- Maximum threads set [Not completed]
- Cache clearing [Not completed]

The performance is evaluated using the following parameters of interest:

- Real time: wall-clock
- User time: CPU time spent executing process
- Sys time: CPU time spent in kernel/OS
- Idle time: $t_{\text{wait}} = t_{\text{real}} - (t_{\text{user}} + t_{\text{sys}})$, where reductionist to receive, respond and talk to disk
- CPU of process: $\frac{t_{\text{user}}+t_{\text{sys}}}{t_{\text{real}}} \times 100$ [%]
- Memory usage: from max. resident set size: $\frac{\text{RES}}{1024^2}$ [MB]

Scripts and data can be found in the following Git repository.
Tests are completed with the following versions:

Name	Version	Build	Channel
`pyfive`	1.1.1	pyhd8ed1ab_2	conda-forge
`pyactivestorage`	0.7.0.dev2+gf4516e6a8	pypi_0	pypi

Table 1: Version Information on libraries

2 Comparing PyActiveStorage with respect to Local Storage

The execution time of PyActiveStorage to perform data reductions are compared to reductions performed in local storage. For a given file and dataset tested, the proportion of the file tested is varied by so called target percentages. This allows for one to observe how the reductions scales with the percentage of the sliced file. The first results are completed on a file with 3400 chunks and the second file the same as the first file but with a coarser chunking of 64 chunks. Details on the dataset and respective files are presented in Table ??. The following tests are performed with datasets accessed via the `https` and `s3` protocols to also compare how this impacts the execution time.

File	`ch330a.pc19790301-bnl.nc`	`ch330a.pc19790301-def.nc`
Filesize	17.96 GB	17.81 GB
Dataset	`UM_m01s16i202_vn1106`	`UM_m01s16i202_vn1106`
Shape	(40, 1920, 2560)	(40, 1920, 2560)
Chunk shape	(4, 113, 128)	(10, 480, 640)
Chunks per axis	[10, 17, 20]	[4, 4, 4]
Number of chunks	3400	64

Table 2: Meta information about the `ch330a.pc19790301-bnl.nc` and `ch330a.pc19790301-def.nc` files for the `UM_m01s16i202_vn1106` dataset.

The target slicing of the file is shown in Table ??, with the exact slicing applied to each axis shown along side the exact percentage of the slicing performed on the file.

Target (%)	Slice Range	Actual (%)
100	[0:40, 0:1920, 0:2560]	100
75	[0:36, 0:1753, 0:2337]	74.95
50	[0:31, 0:1525, 0:2032]	48.74
25	[0:25, 0:1209, 0:1612]	24.78
10	[0:18, 0:891, 0:1188]	9.70
5	[0:14, 0:707, 0:942]	4.74
1	[0:8, 0:413, 0:551]	0.926

Table 3: The `ch330a.pc19790301-bnl.nc` file is tested at varying target file percentage for the and `UM_m01s16i202_vn1106` dataset, where this file has 3400 chunks. The slicing applied to achieve the target are shown with the actual percentage this represents.

The results are obtained by running the same reduction multiple times and averaging over the results in order to account for fluctuations in the network speed. Due to the time taken to run over large slices, fewer iterations are ran for the larger target percentage tests. The results shown in Figure ?? for the 3400 chunked file and in Figure ?? for the 64 chunk file. It is clear when comparing both figures that having fewer chunks allows for data reductions to be performed faster. In both cases the reductions done using PyActiveStorage are significantly faster, especially when looking at the ratio in the execution time for Local with respect to Active storage. The ratio grows with the target file size, demonstrating the advantage to performing reductions with the PyActiveStorage tool. For completeness Tables ?? and ?? list the average execution times and the number of iterations ran for the varying target sizes. The ratios between Local and Active storage for varying target slices are shown in Figure ?? for visualisation. The ratio between the two files execution time for a given storage type, are visualised in Figure ??, showing that the active storage is independent of the target size, whereas for local storage, this ratio increases with target size.

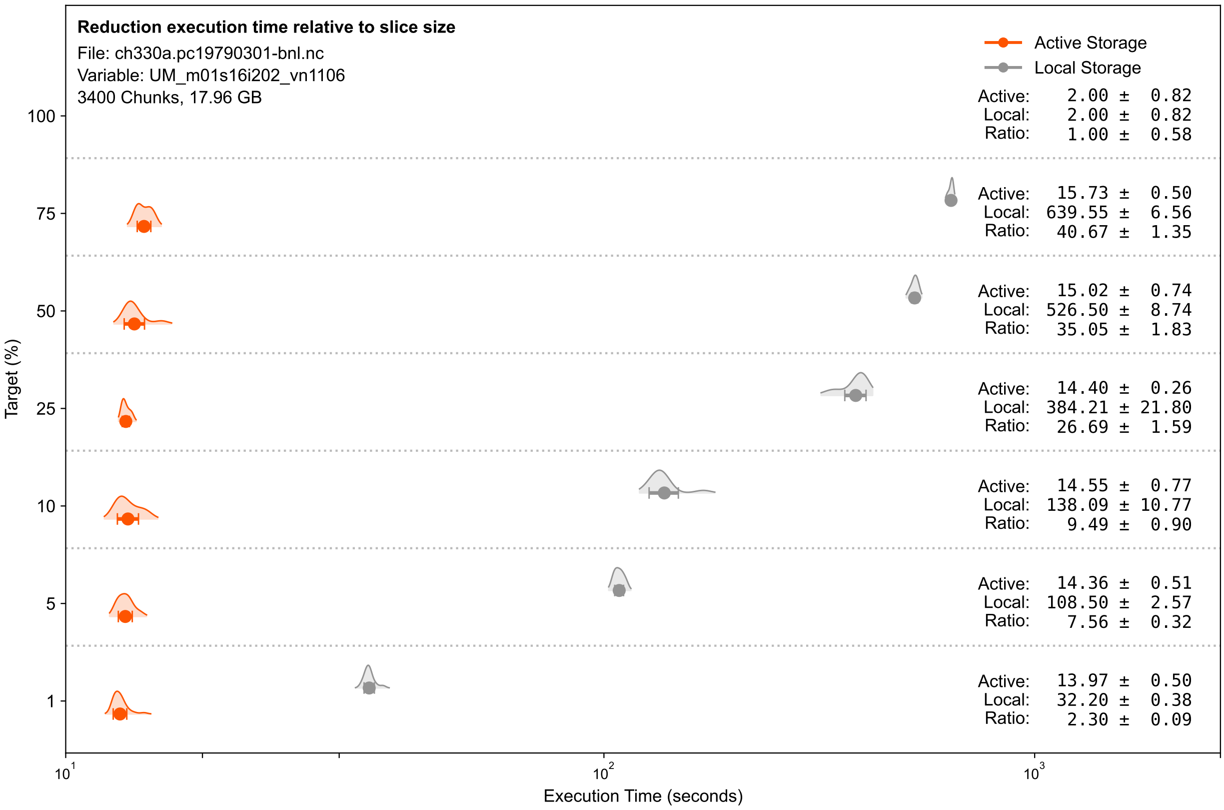


Figure 1: Execution time results of performing data reductions on the file `ch330a.pc19790301-bnl.nc` which has 3400 chunks. Reductions on the dataset in Active storage compared to reductions completed in local storage. Values shown for varying Target percentage of the file, with the average execution time for local storage, active storage, and the ratio of Local/Active shown on the right hand side.

	nIter		Execution Time (s)		
Target (%)	Active	Local	Active	Local	Local/Active
100	.	.	.	.	.
75	5	5	15.73 ± 6.56	639.55 ± 0.50	40.67 ± 1.35
50	5	5	15.02 ± 8.74	526.50 ± 0.74	35.05 ± 1.83
25	5	5	14.40 ± 21.80	384.21 ± 0.26	26.69 ± 1.59
10	10	10	14.55 ± 10.77	138.09 ± 0.77	9.49 ± 0.90
5	10	10	14.92 ± 2.57	108.50 ± 1.74	7.27 ± 0.87
1	30	30	13.97 ± 0.38	32.20 ± 0.50	2.30 ± 0.09

Table 4: The number of iterations applied to measure the execution time when running with PyActiveStorage or Local storage are provided. Results of the execution time are provided for both tests and the ratio between the Local and PyActiveStorage performance.

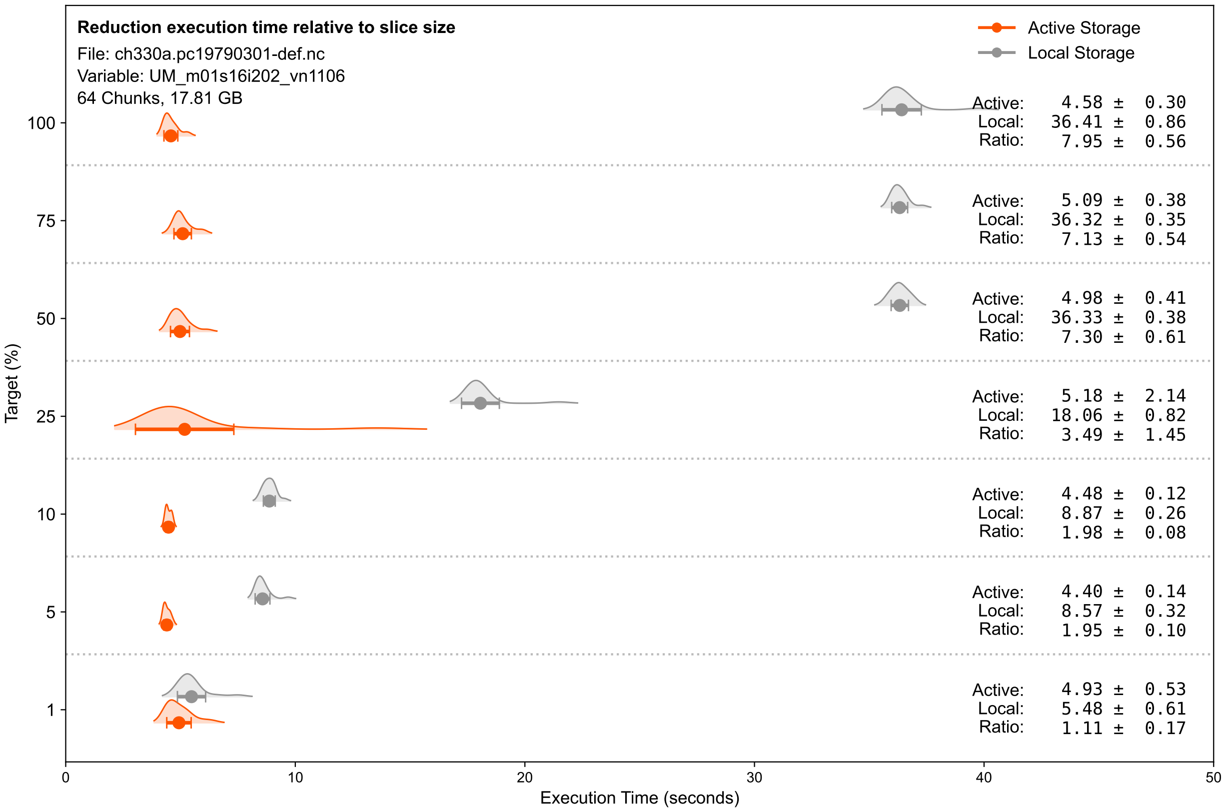
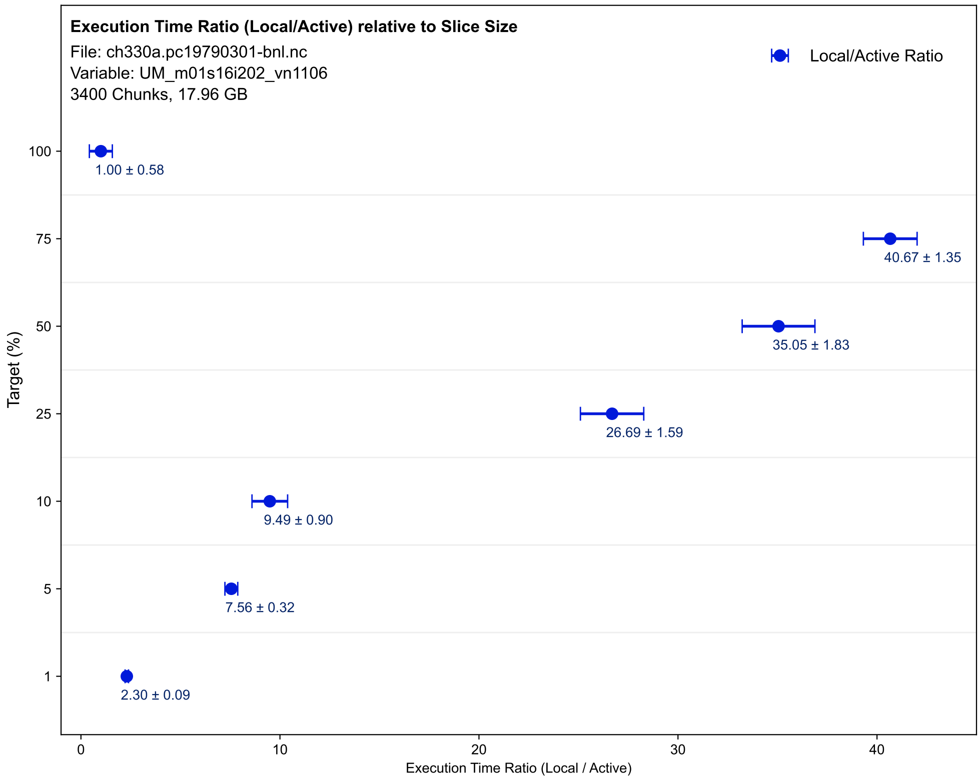


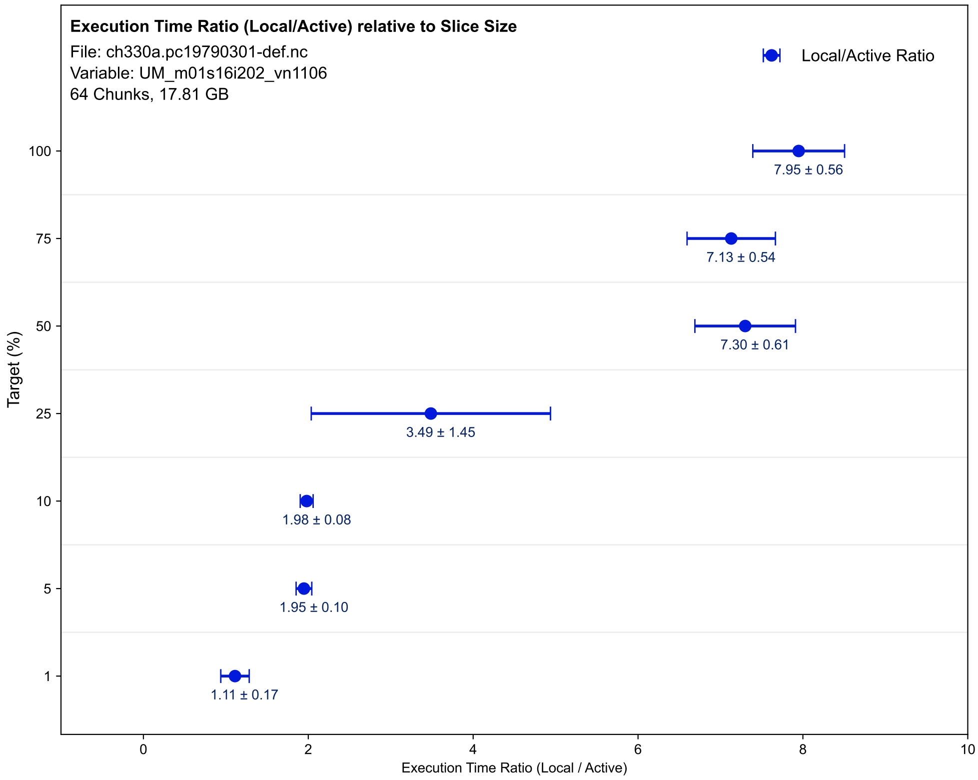
Figure 2: Execution time results of performing data reductions on the file `ch330a.pc19790301-def.nc` which has 64 chunks. Reductions on the dataset in Active storage compared to reductions completed in local storage. Values shown for varying Target percentage of the file, with the average execution time for local storage, active storage, and the ratio of Local/Active shown on the right hand side.

	nIter		Execution Time (s)		
Target (%)	Active	Local	Active	Local	Local/Active
100	19	19	4.58 ± 0.30	36.41 ± 0.86	7.95 ± 0.56
75	19	19	5.09 ± 0.38	36.32 ± 0.35	7.13 ± 0.54
50	19	19	4.98 ± 0.41	36.33 ± 0.38	7.30 ± 0.61
25	19	19	5.18 ± 2.14	18.06 ± 0.82	3.49 ± 1.45
10	19	19	4.48 ± 0.12	8.87 ± 0.26	1.98 ± 0.08
5	19	19	4.40 ± 0.14	8.57 ± 0.32	1.95 ± 0.10
1	19	19	4.93 ± 0.53	5.48 ± 0.61	1.11 ± 0.17

Table 5: The number of iterations applied to measure the execution time when running with PyActiveStorage or Local storage are provided. Results of the execution time are provided for both tests and the ratio between the Local and PyActiveStorage performance.

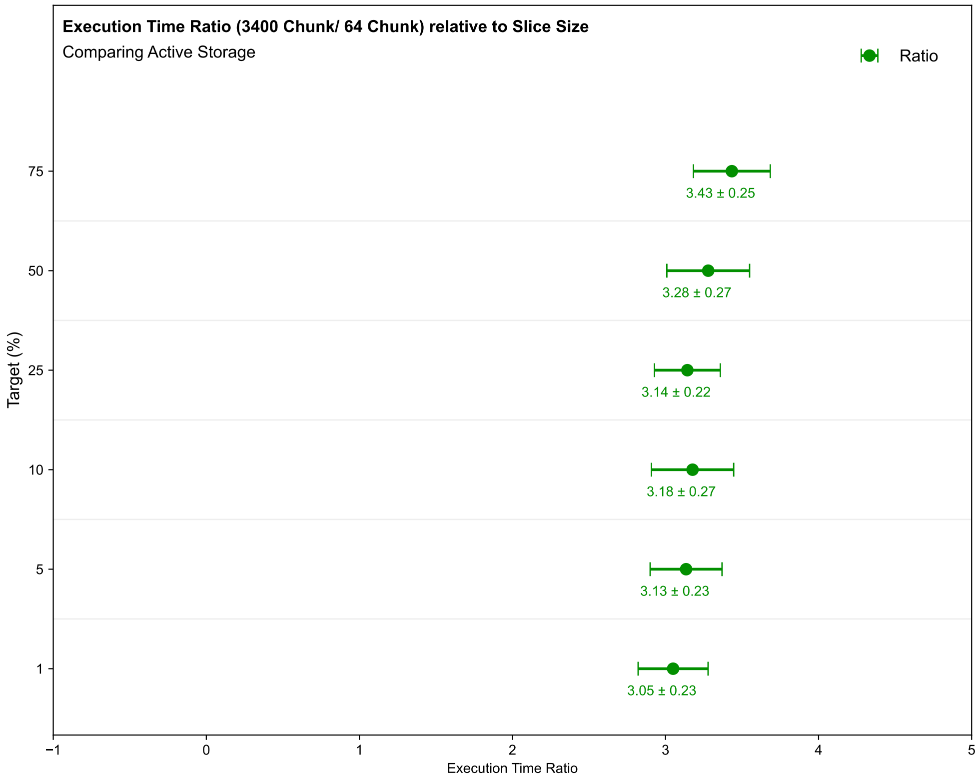


(a) File of 3400 chunks

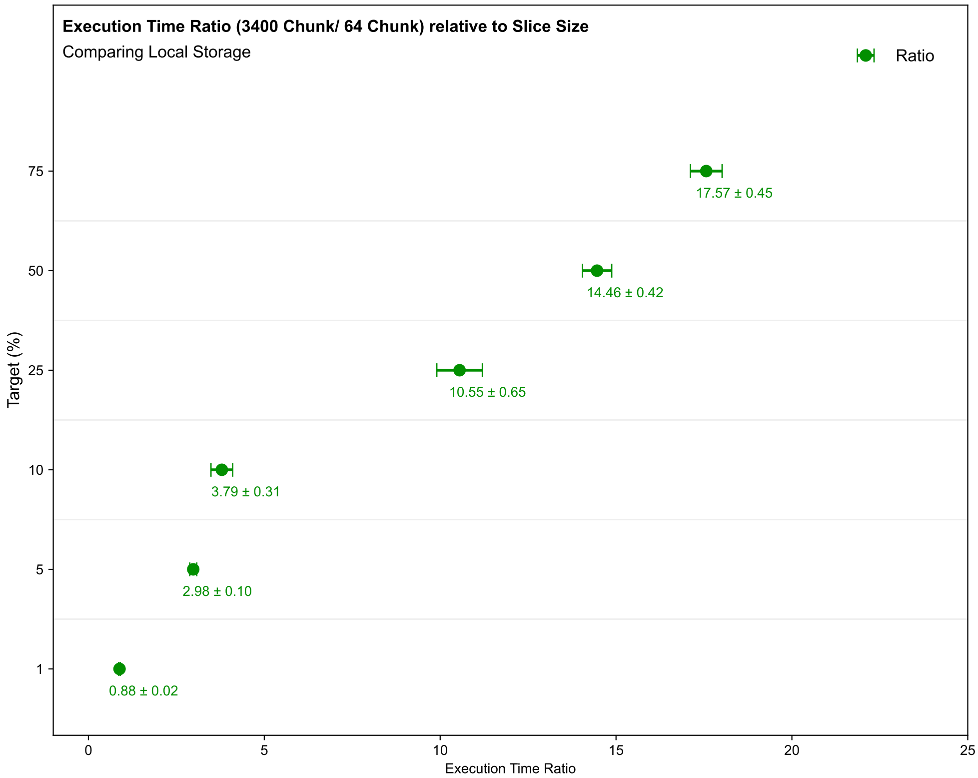


(b) File of 64 chunks

Figure 3: Ratio of execution time comparing local and active storage for the same file and dataset but with a different chunking structure.



(a) Active Storage



(b) Local Storage

Figure 4: Ratio of execution time comparing different chunking, for a given storage setting.

3 Comparing https and s3 protocols

For the file and dataset presented in Table ??, a test is performed to understand how the different protocols to retrieve the file varies the performance of the data reductions. The results of the study are presented in Figure ??. For both protocols, the active storage executes reductions faster. The two protocols are compatible when considering the use of active storage.

File	`cl_Amon_UKESM1-0-LL_ssp370SST-lowNTCF_r1i1p1f2_gn_205001-209912.nc`
File size	2.36 GB
Dataset	`cl`
Shape	(600, 85, 144, 192)
Chunk shape	(1, 43, 72, 96)
Chunks per axis	(600, 2, 2, 2)
Total number of chunks	4800

Table 6

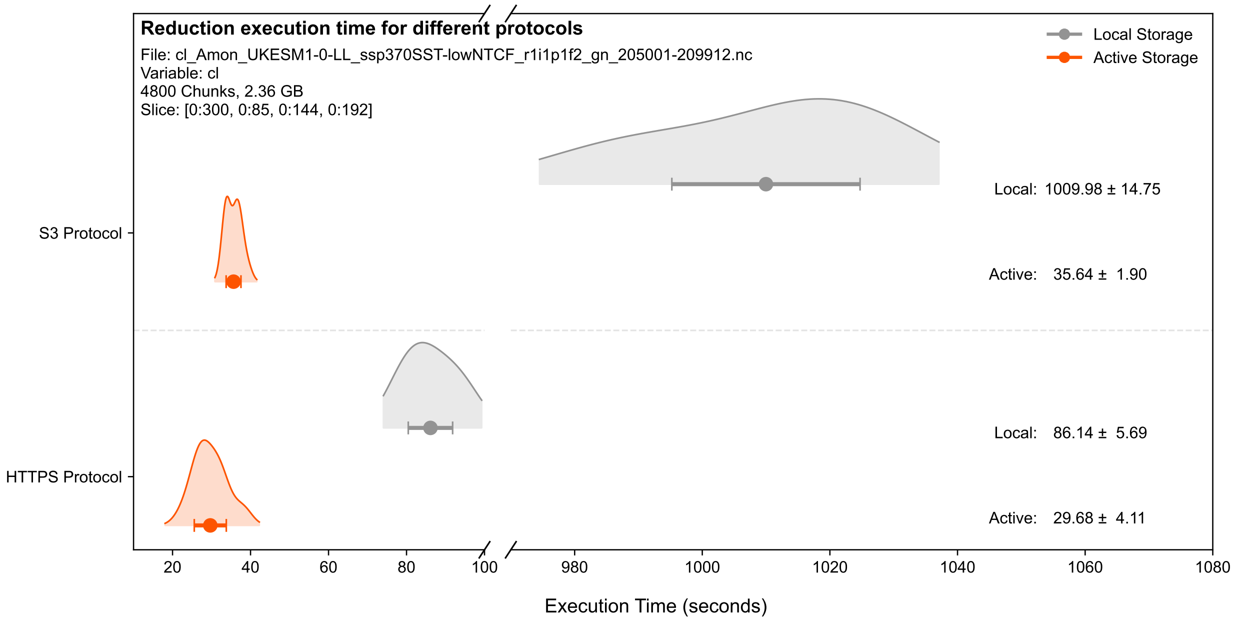


Figure 5

4 Caching Tests

Dataset Size	State	Wait Time (s)	CPU (%)	Memory (MB)
3,400 Chunks (5%)	Cold	16.28	18.36	316.31
3,400 Chunks (5%)	Warm	11.35	20.93	311.78
64 Chunks (100%)	Cold	7.96	18.02	306.48
64 Chunks (100%)	Warm	3.29	28.31	301.80

Table 7: Comparison of Remote Calculation KPIs: Cold vs. Warm Cache

5 Test Conditions & Results

In order to understand the performance, multiple iterations of a given test are completed in order to fill histograms and evaluate the average performance. Caching of the data will directly influence its performance. Therefore, an initial "cold" run is ran such that for the following "warm" runs, the caching of the data is already present on the Reductionist. Typically 60 warm runs are recorded, however if a process was very slow, it was capped at an earlier number of iterations. This will be indicated in the respective testing sections. Unless stated otherwise, tests are performed with the number of maximum threads is set to 100, the method applied is `min` and the data file is retrieved from the `s3` object store.

5.1 Test 1

	Test 1
File	`ch330a.pc19790301-bnl.nc` (17.96 GB)
Dataset	`UM_m01s16i202_vn1106`
Shape	(40, 1920, 2560)
Chunk shape	(4, 113, 128)
Chunks per axis	[10, 17, 20]
Total number of chunks	3400
Slicing applied	[0:4, 0:30, 0:30]
No. of Iterations	60

Table 8

Results	Test 1
Real Time [s]	14.696 ± 1.842
User Time [s]	1.610 ± 0.445
Sys Time [s]	0.902 ± 0.232
Wait Time [s]	12.185 ± 1.189
Memory [MB]	286.142 ± 10.856
CPU Util [%]	16.904 ± 1.700

Table 9: Test 1 Results

5.2 Test 2

	Test 2
File	`ch330a.pc19790301-bnl.nc` (17.96 GB)
Dataset	`UM_m01s30i202_vn1106`
Shape	(40, 7, 1921, 2560)
Chunk shape	(4, 1, 113, 128)
Chunks per axis	[10, 7, 17, 20]
Total number of chunks	23800
Slicing applied	[0:4, 0:1, 0:30, 0:30]
No. of Iterations	20

Table 10

Results	Test 2
Real Time [s]	65.147 ± 1.622
User Time [s]	6.154 ± 0.184
Sys Time [s]	3.503 ± 0.095
Wait Time [s]	55.489 ± 1.380
Memory [MB]	304.298 ± 8.933
CPU Util [%]	14.824 ± 0.178

Table 11: Test 2 Results

5.3 Test 3

	Test 3
File	`ch330a.pc19790301-def.nc` (17.81 GB)
Dataset	`UM_m01s16i202_vn1106`
Shape	(40, 1920, 2560)
Chunk shape	(10, 480, 640)
Chunks per axis	[4, 4, 4]
Total number of chunks	64
Slicing applied	[0:4, 0:30, 0:30]
No. of Iterations	60

Table 12

Results	Test 3
Real Time [s]	4.937 ± 1.283
User Time [s]	0.657 ± 0.219
Sys Time [s]	0.345 ± 0.107
Wait Time [s]	3.935 ± 0.985
Memory [MB]	269.209 ± 9.498
CPU Util [%]	20.136 ± 1.744

Table 13: Test 3 Results

5.4 Test 4

	Test 4
File	`ch330a.pc19790301-def.nc` (17.81 GB)
Dataset	`UM_m01s30i202_vn1106`
Shape	(40, 7, 1921, 2560)
Chunk shape	(8, 1, 385, 512)
Chunks per axis	[5, 7, 5, 5]
Total number of chunks	875
Slicing applied	0:4, 0:1, 0:30, 0:30]
No. of Iterations	20

Table 14

Results	Test 4
Real Time [s]	18.605 ± 0.816
User Time [s]	1.876 ± 0.135
Sys Time [s]	1.058 ± 0.054
Wait Time [s]	15.673 ± 0.654
Memory [MB]	286.00 ± 9.291
CPU Util [%]	15.756 ± 0.463

Table 15: Test 4 Results

5.5 Test 5

	Test 5
File	`cl_Amon_UKESM1-0-LL_ssp370SST -lowNTCF_r1i1p1f2_gn_205001-209912.nc` (2.36 GB)
Dataset	`cl`
Shape	(600, 85, 144, 192)
Chunk shape	(1, 43, 72, 96)
Chunks per axis	(600, 2, 2, 2)
Total number of chunks	4800
Slicing applied	[0:4, 0:1, 0:30, 0:30]
No. of Iterations	20

Table 16

Results	Test 5
Real Time [s]	30.014 ± 0.898
User Time [s]	2.981 ± 0.098
Sys Time [s]	1.606 ± 0.073
Wait Time [s]	25.428 ± 0.772
Memory [MB]	289.440 ± 8.367
CPU Util [%]	15.282 ± 0.266

Table 17: Test 5 Results

5.6 Test 6

	Test 6
File	`da193a_25_day__198808-198808.nc`
Dataset	`m01s06i247_4`
Shape	(30, 39, 325, 432)
Chunk shape	(1, 39, 325, 432)
Chunks per axis	(30, 1, 1, 1))
Total number of chunks	30
Slicing applied	[0:4, 0:1, 0:30, 0:30]
No. of Iterations	60

Table 18

Results	Test 6
Real Time [s]	1.242 ± 0.091
User Time [s]	0.333 ± 0.012
Sys Time [s]	0.138 ± 0.008
Wait Time [s]	0.772 ± 0.087
Memory [MB]	206.392 ± 1.525
CPU Util [%]	38.032 ± 2.294

Table 19: Test 6 Results

5.7 Test 7

	Test 7
File	`da193a_25_6hr_t_pt_cordex__198807-198807.nc` (6.13 GB)
Dataset	`m01s30i111`
Shape	(120, 85, 324, 432)
Chunk shape	(1, 85, 324, 432)
Chunks per axis	(120, 1, 1, 1))
Total number of chunks	120
Slicing applied	[0:4, 0:1, 0:30, 0:30]
No. of Iterations	20

Table 20

Results	Test 7
Real Time [s]	2.917 ± 0.167
User Time [s]	0.497 ± 0.052
Sys Time [s]	0.232 ± 0.031
Wait Time [s]	2.187 ± 0.106
Memory [MB]	276.494 ± 8.837
CPU Util [%]	24.957 ± 1.729

Table 21: Test 7 Results

5.8 Test 8

Test 8 has the same conditions as Test 1 except now it is ran on the UoR Eduroam network instead in order to compare the difference between the network and wifi performance of the PyActiveStorage. As running on the WiFi is much slower, only 10 iterations are ran to determine the results.

Results	Test 8
Real Time [s]	80.063 ± 18.940
User Time [s]	6.533 ± 0.672
Sys Time [s]	3.132 ± 0.390
Wait Time [s]	70.398 ± 19.189
Memory [MB]	290.321 ± 5.774
CPU Util [%]	12.533 ± 2.230

Table 22: Test 8 Results

5.9 Test 9

Test 9 is the same as Test 5 except in how the data is read in. Test 5 uses the s3 object store, with the file stored in the bnl bucket. For Test 9, the same file is accessed via https of the ESGF CEDA catalogue .

Results	Test 9
Real Time [s]	140.162 ± 9.966
User Time [s]	3.594 ± 0.299
Sys Time [s]	1.704 ± 0.111
Wait Time [s]	134.864 ± 9.688
Memory [MB]	168.608 ± 14.878
CPU Util [%]	3.785 ± 0.212

Table 23: Test 9 Results

5.10 Test 10

Results	Test 10
Real Time [s]	16.346 ± 0.619
User Time [s]	1.756 ± 0.122
Sys Time [s]	0.971 ± 0.069
Wait Time [s]	13.620 ± 0.518
Memory [MB]	333.869 ± 8.949
CPU Util [%]	16.675 ± 0.867

Table 24: Test 10 Results

6 Tests Rough draft, need to check still

6.1 File sizes, data shapes and chunking variations

6.1.1 Data Shape

This test aims to understand how the choice of a HDF5 dataset, i.e its shape, influences the performance. Shown in Table **??** are the difference in the datasets chosen. The additional dimension in the

UM_m01s30i202_vn1106 dataset slows down the running time and therefore it is tested on a smaller number of iterations.

	Test 1	Test 2
File	ch330a.pc19790301-bnl.nc (17.96 GB)	
Dataset	UM_m01s16i202_vn1106	UM_m01s30i202_vn1106
Shape	(40, 1920, 2560)	(40, 7, 1921, 2560)
Chunk shape	(4, 113, 128)	(4, 1, 113, 128)
Chunks per axis	[10, 17, 20]	[10, 7, 17, 20]
Total number of chunks	3400	23800
Slicing applied	[0:4, 0:30, 0:30]	[0:4, 0:1, 0:30, 0:30]
No. of Iterations	60	20

Table 25: Properties of the tests performed when varying the HDF5 Dataset in the ch330a.pc19790301-bnl.nc file. For both tests the number of maximum threads is set to 100, the method applied is min and the data file is retrieved from the s3 object store.

Results	Test 1	Test 2	$T1/T2$
Real Time [s]	14.696 ± 1.842	65.147 ± 1.622	0.226 ± 0.029
User Time [s]	1.610 ± 0.445	6.154 ± 0.184	0.262 ± 0.073
Sys Time [s]	0.902 ± 0.232	3.503 ± 0.095	0.257 ± 0.067
Wait Time [s]	12.185 ± 1.189	55.489 ± 1.380	0.220 ± 0.022
Memory [MB]	286.142 ± 10.856	304.298 ± 8.933	0.940 ± 0.046
CPU Util [%]	16.904 ± 1.700	14.824 ± 0.178	1.140 ± 0.116

Table 26: Results obtained from Test 1 and 2, showing all parameters of interest. The ratio between the two tests, $T1/T2$ is shown with propagated uncertainties.

Table **??** shows the averaged results for the parameters of interest. Main observations are as follows:

- CPU and Memory values are compatible between the tests
- Key difference in the Wait Time (I/O of the data). Although only one additional chunk is selected, the variable has seven times more chunks in the file which will increase the overhead due to factors such as additional metadata lookup and chunk indexing.

6.1.2 Coarser Chunking

The same HDF5 datasets are studied in another file where the number of chunks is coarser. The number of chunks along with the testing conditions can be found in Table **??**.

	Test 3	Test 4
File	ch330a.pc19790301-def.nc (17.81 GB)	
Dataset	UM_m01s16i202_vn1106	UM_m01s30i202_vn1106
Shape	(40, 1920, 2560)	(40, 7, 1921, 2560)
Chunk shape	(10, 480, 640)	(8, 1, 385, 512)
Chunks per axis	[4, 4, 4]	[5, 7, 5, 5]
Total number of chunks	64	875
Slicing applied	[0:4, 0:30, 0:30]	[0:4, 0:1, 0:30, 0:30]
No. of Iterations	60	20

Table 27: Variables compared from the ch330a.pc19790301-def.nc file, for a maximum number of threads of 100, method applied is min and the data file is retrieved from the s3 object store.

Results	Test 3	Test 4	$T3/T4$
Real Time [s]	4.937 ± 1.283	18.605 ± 0.816	0.265 ± 0.070
User Time [s]	0.657 ± 0.219	1.876 ± 0.135	0.350 ± 0.118
Sys Time [s]	0.345 ± 0.107	1.058 ± 0.054	0.326 ± 0.103
Wait Time [s]	3.935 ± 0.985	15.673 ± 0.654	0.251 ± 0.064
Memory [MB]	269.209 ± 9.498	286.00 ± 9.291	0.941 ± 0.047
CPU Util [%]	20.136 ± 1.744	15.756 ± 0.463	1.278 ± 0.115

Table 28: Variables compared from the `ch330a.pc19790301-def.nc` file. The factor $T3/T4$ is shown with propagated uncertainty based on the relative standard deviations of both tests.

Table ?? shows the averaged results for the parameters of interest. Main observations are as follows:

- The ratios for all parameters are very similar to that observed in Table ??, showing that the additional dimension in the data shape has the same impact, even if other axis are chunked coarser.

Results	Test 1	Test 3
Real Time [s]	14.696 ± 1.842	4.937 ± 1.283
User Time [s]	1.610 ± 0.445	0.657 ± 0.219
Sys Time [s]	0.902 ± 0.232	0.345 ± 0.107
Wait Time [s]	12.185 ± 1.189	3.935 ± 0.985
Memory [MB]	286.142 ± 10.856	269.209 ± 9.498
CPU Util [%]	16.904 ± 1.700	20.136 ± 1.744
Results	Test 2	Test 4
Real Time [s]	65.147 ± 1.622	18.605 ± 0.816
User Time [s]	6.154 ± 0.184	1.876 ± 0.135
Sys Time [s]	3.503 ± 0.095	1.058 ± 0.054
Wait Time [s]	55.489 ± 1.380	15.673 ± 0.654
Memory [MB]	304.298 ± 8.933	286.00 ± 9.291
CPU Util [%]	14.824 ± 0.178	15.756 ± 0.463

Table 29: Variables compared from the `ch330a.pc19790301-def.nc` file, for a maximum number of threads of 100, method applied is `min` and the data file is retrieved from the `s3` object store.

The impact of the coarser chunking can be compared for the same dataset between the two files. The results for how the chunking impacts the parameters of interest are shown in Table ?? where Test 1 and 3 compare for the `UM_m01s16i202_vn1106` data set, and Test 2 and 4 the `UM_m01s30i202_vn1106` data set. The main observations are as follows:

- The smaller number of chunks allows for the reductions to be completed faster.
- The CPU usuage is similar due to a short process time, there are fewer idle resources.
- T2/T4 ratio takes more time due to

6.2 UoR Ethernet vs Eduroam

6.3 s3 v https